Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

XGBoost XAI study notebook

This notebook is a model-specific XAI study for the trained `view_e_full` XGBoost pipeline reconstructed from `churn_analysis_helpers.py`.

It is intentionally organised around the model's function class rather than only around predictive replay.

Study goals

We treat the fitted model as

$$f_{\text{xgb}}(x) = \sum_{m=1}^{M} \eta T_m(x), \qquad p_{\text{xgb}}(x) = \sigma\big(f_{\text{xgb}}(x)\big)\,.$$

Because each tree induces axis-aligned splits, the fitted logit surface is piecewise constant inside leaf regions. The central questions are therefore:

1. Which engineered features dominate global risk allocation?
2. Which interactions create the sharpest regime changes?
3. Where does the model become overconfident or brittle?
4. What do its conditional decision surfaces actually look like?

Required files in `./`

- `train.csv`
- `test.csv`
- `WA_Fn-UseC_-Telco-Customer-Churn.csv`
- `xgb_view_e_full_final_model.json`
- `xgb_view_e_full_feature_columns.json`
- `xgb_view_e_full_training_metadata.json`
- `churn_analysis_helpers.py`

In [1]:
```
from pathlib import Path
import json
import warnings
```

```
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import shap
import xgboost as xgb

from sklearn.metrics import roc_auc_score, brier_score_loss, log_lo
from sklearn.inspection import PartialDependenceDisplay
from sklearn.calibration import calibration_curve

from churn_analysis_helpers import (
    CFG,
    seed_everything,
    setup_plotting,
    ensure_output_dir,
    savefig,
    build_feature_frames,
    build_xgb_runtime_frames,
    load_xgb_artifacts,
    align_xgb_features,
    pred_summary,
    make_decile_frame,
    cumulative_capture,
    compute_xgb_shap_values,
)

seed_everything(67)
setup_plotting()

cfg = CFG()
ensure_output_dir(cfg)
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

1. Reconstruct the exact XGBoost runtime feature space

This section does not create a new feature pipeline. It uses the helper exactly as written, rebuilds the runtime frame, loads the saved model artefacts, aligns the runtime columns to the saved feature column order, and obtains fitted probabilities on train and test.

In [2]:
```
bundle = build_feature_frames(cfg)
runtime = build_xgb_runtime_frames(bundle)
```

```
xgb_model, feature_cols_saved, xgb_metadata = load_xgb_artifacts(cf

X_train_aligned, extra_train_cols = align_xgb_features(runtime["X_t
X_test_aligned, extra_test_cols = align_xgb_features(runtime["X_tes

xgb_train_pred = xgb_model.predict_proba(X_train_aligned)[:, 1]
xgb_test_pred = xgb_model.predict_proba(X_test_aligned)[:, 1]

n_comp = len(bundle["y"])
comp_mask = np.arange(len(X_train_aligned)) < n_comp

xgb_comp_train_pred = xgb_train_pred[comp_mask]
y = bundle["y"]

print("Saved feature columns:", len(feature_cols_saved))
print("Runtime aligned train shape:", X_train_aligned.shape)
print("Runtime aligned test shape :", X_test_aligned.shape)
print("Competition-train AUC:", f"{roc_auc_score(y, xgb_comp_train_
print("Competition-train log loss:", f"{log_loss(y, xgb_comp_train_
print("Competition-train Brier:", f"{brier_score_loss(y, xgb_comp_t
print("Extra runtime cols ignored:", len(extra_train_cols))
display(pd.Series(pred_summary(xgb_comp_train_pred, "xgb_train")).t
```

```
Saved feature columns: 161
Runtime aligned train shape: (601237, 161)
Runtime aligned test shape : (254655, 161)
Competition-train AUC: 0.933437
Competition-train log loss: 0.268055
Competition-train Brier: 0.085993
Extra runtime cols ignored: 15
```

	value
xgb_train_min	0.000003
xgb_train_p01	0.000015
xgb_train_p05	0.000050
xgb_train_median	0.053924
xgb_train_mean	0.225706
xgb_train_p95	0.840071
xgb_train_p99	0.958884
xgb_train_max	0.999883

In [3]:
```
analysis_train = pd.DataFrame({
    cfg.id_col: bundle["train"][cfg.id_col].values,
    "y_true": y,
    "xgb_pred": xgb_comp_train_pred,
})

analysis_test = pd.DataFrame({
    cfg.id_col: bundle["test"][cfg.id_col].values,
```

```
    "xgb_pred": xgb_test_pred,
})

analysis_train["pred_rank_pct"] = analysis_train["xgb_pred"].rank(p
analysis_train["error"] = analysis_train["y_true"] - analysis_train
analysis_train["abs_error"] = analysis_train["error"].abs()
analysis_train["hard_label"] = (analysis_train["xgb_pred"] >= 0.5).
analysis_train["case_type"] = np.select(
    [
        (analysis_train["y_true"] == 1) & (analysis_train["hard_lab
        (analysis_train["y_true"] == 0) & (analysis_train["hard_lab
        (analysis_train["y_true"] == 0) & (analysis_train["hard_lab
        (analysis_train["y_true"] == 1) & (analysis_train["hard_lab
    ],
    ["TP", "TN", "FP", "FN"],
    default="other"
)

analysis_train["risk_band"] = pd.qcut(
    analysis_train["xgb_pred"],
    q=5,
    labels=["very_low", "low", "mid", "high", "very_high"],
    duplicates="drop",
)

analysis_train.head()
```

Out[3]:

	id	y_true	xgb_pred	pred_rank_pct	error	abs_error	hard_label	c
0	0	0	0.000078	0.078251	-0.000078	0.000078	0	
1	1	0	0.001616	0.197323	-0.001616	0.001616	0	
2	2	0	0.324847	0.705830	-0.324847	0.324847	0	
3	3	1	0.683729	0.878599	0.316271	0.316271	1	
4	4	1	0.775512	0.921608	0.224488	0.224488	1	

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

2. Predictive and calibration diagnostics

This is the behavioural layer of the study.

We inspect:

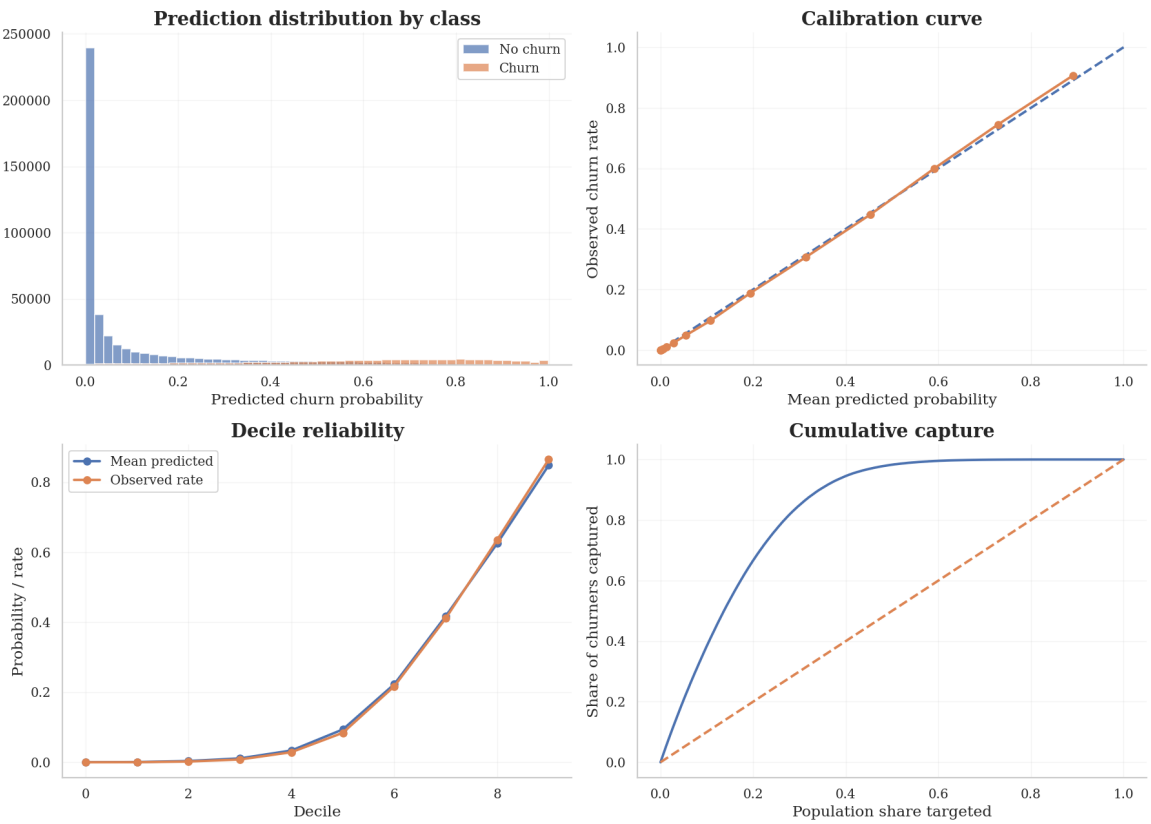
- score distribution,

- empirical reliability,
- decile-level calibration,
- cumulative capture,
- error concentration in the tails.

If the model is exploiting sharp tree partitions well, high-risk bins should produce strong capture and the top tail should contain a large share of positive churn cases.

In [4]:
```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

axes[0, 0].hist(analysis_train.loc[analysis_train["y_true"] == 0, "
axes[0, 0].hist(analysis_train.loc[analysis_train["y_true"] == 1, "
axes[0, 0].set_title("Prediction distribution by class")
axes[0, 0].set_xlabel("Predicted churn probability")
axes[0, 0].legend()

prob_true, prob_pred = calibration_curve(analysis_train["y_true"],
axes[0, 1].plot([0, 1], [0, 1], linestyle="--")
axes[0, 1].plot(prob_pred, prob_true, marker="o")
axes[0, 1].set_title("Calibration curve")
axes[0, 1].set_xlabel("Mean predicted probability")
axes[0, 1].set_ylabel("Observed churn rate")

dec = make_decile_frame(analysis_train["y_true"], analysis_train["x
axes[1, 0].plot(dec["mean_pred"], marker="o", label="Mean predicted
axes[1, 0].plot(dec["churn_rate"], marker="o", label="Observed rate
axes[1, 0].set_title("Decile reliability")
axes[1, 0].set_xlabel("Decile")
axes[1, 0].set_ylabel("Probability / rate")
axes[1, 0].legend()

cap = cumulative_capture(analysis_train["y_true"], analysis_train["
axes[1, 1].plot(cap["population_share"], cap["capture_rate"])
axes[1, 1].plot([0, 1], [0, 1], linestyle="--")
axes[1, 1].set_title("Cumulative capture")
axes[1, 1].set_xlabel("Population share targeted")
axes[1, 1].set_ylabel("Share of churners captured")

plt.tight_layout()
savefig(cfg, "xgb_01_core_diagnostics.png")
plt.show()

display(dec)
```

```
saved: analysis_outputs\xgb_01_core_diagnostics.png
```



	bin	mean_pred	churn_rate	n	pred_min	pred_max
0	0	0.000052	0.000000	59420	0.000003	0.000110
1	1	0.000682	0.000236	59419	0.000110	0.001696
2	2	0.003563	0.002003	59419	0.001696	0.005920
3	3	0.011040	0.007927	59420	0.005920	0.018740
4	4	0.033567	0.028728	59419	0.018741	0.053924
5	5	0.093750	0.084165	59419	0.053925	0.145461
6	6	0.223051	0.217115	59420	0.145461	0.313004
7	7	0.417723	0.411215	59419	0.313006	0.523134
8	8	0.625240	0.635672	59419	0.523141	0.727878
9	9	0.848388	0.865012	59420	0.727878	0.999883

In [5]:
```
tail_summary = pd.DataFrame({
    "segment": ["Top 1%", "Top 5%", "Top 10%", "Bottom 10%"],
    "observed_churn_rate": [
        analysis_train.loc[analysis_train["pred_rank_pct"] >= 0.99,
        analysis_train.loc[analysis_train["pred_rank_pct"] >= 0.95,
        analysis_train.loc[analysis_train["pred_rank_pct"] >= 0.90,
        analysis_train.loc[analysis_train["pred_rank_pct"] <= 0.10,
    ],
    "n": [
        int((analysis_train["pred_rank_pct"] >= 0.99).sum()),
        int((analysis_train["pred_rank_pct"] >= 0.95).sum()),
        int((analysis_train["pred_rank_pct"] >= 0.90).sum()),
        int((analysis_train["pred_rank_pct"] <= 0.10).sum()),
```

```
    ],
})
display(tail_summary)
```

	segment	observed_churn_rate	n
0	Top 1%	0.990071	5942
1	Top 5%	0.928879	29710
2	Top 10%	0.865012	59420
3	Bottom 10%	0.000000	59419

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

3. Global SHAP structure

For the tree model, SHAP is the primary global explanation tool. We compute SHAP values on a competition-train sample and study:

- mean absolute contribution magnitude,
- signed contribution direction,
- summary structure across the sample.

For a tree ensemble, SHAP decomposes the fitted logit additively:

$$f_{\text{xgb}}(x) = \phi_0 + \sum_{j=1}^{p} \phi_j(x).$$

Large $|\phi_j(x)|$ means feature j is strongly moving the logit away from baseline for that case.

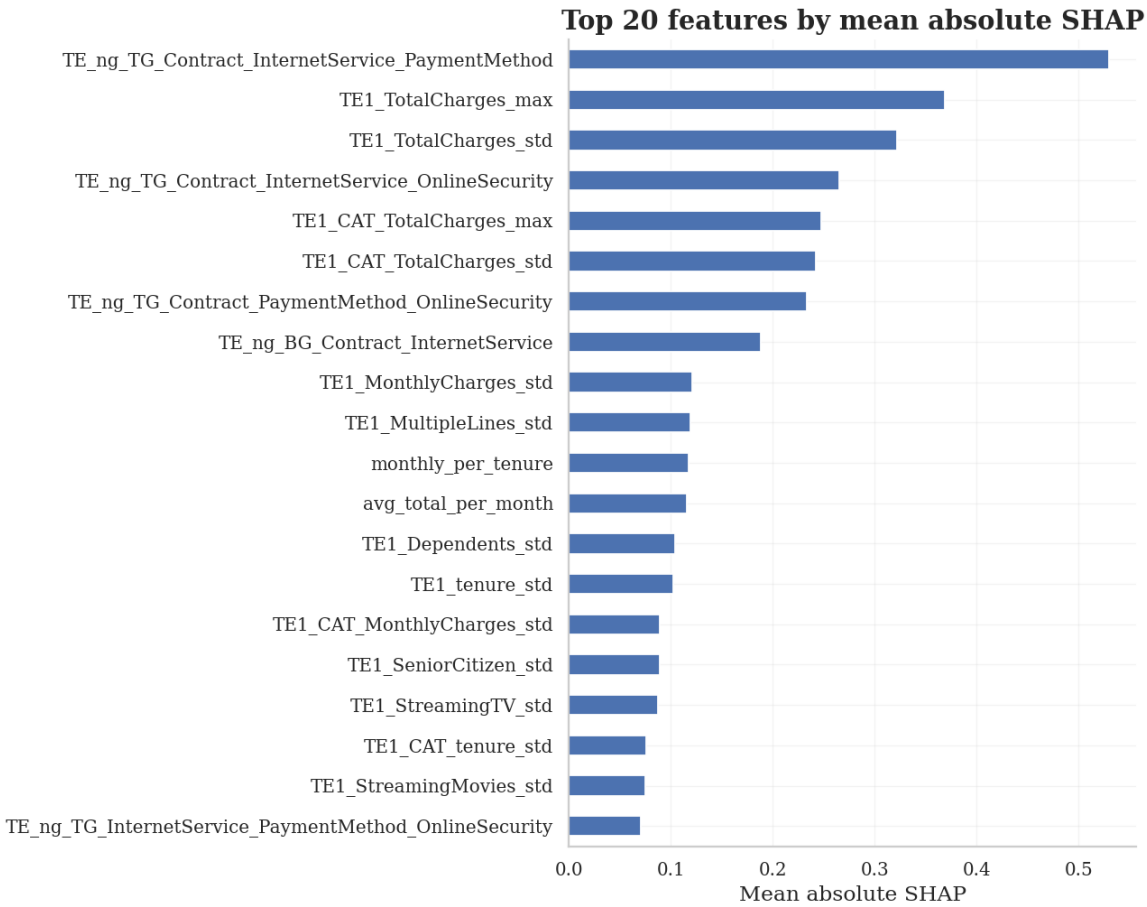
In [6]:
```
shap_idx = np.arange(n_comp)
if len(shap_idx) > cfg.shap_sample_n:
    rng = np.random.default_rng(cfg.seed)
    shap_idx = rng.choice(shap_idx, size=cfg.shap_sample_n, replace

X_shap = X_train_aligned.iloc[shap_idx].copy()
y_shap = y[shap_idx]

explainer, shap_values = compute_xgb_shap_values(xgb_model, X_shap)

expected_value = explainer.expected_value
if isinstance(expected_value, (list, np.ndarray)):
    expected_value_scalar = float(np.array(expected_value).reshape(
```

```
else:
    expected_value_scalar = float(expected_value)

mean_abs_shap = pd.Series(np.abs(shap_values).mean(axis=0), index=X
mean_signed_shap = pd.Series(shap_values.mean(axis=0), index=X_shap

display(mean_abs_shap.head(25).rename("mean_abs_shap").to_frame())
display(mean_signed_shap.head(25).rename("mean_signed_shap").to_fra

plt.figure(figsize=(10, 8))
mean_abs_shap.head(20).sort_values().plot(kind="barh")
plt.title("Top 20 features by mean absolute SHAP")
plt.xlabel("Mean absolute SHAP")
savefig(cfg, "xgb_02_mean_abs_shap.png")
plt.show()

shap.summary_plot(shap_values, X_shap, max_display=20, show=False)
savefig(cfg, "xgb_03_shap_beeswarm.png")
plt.show()
```

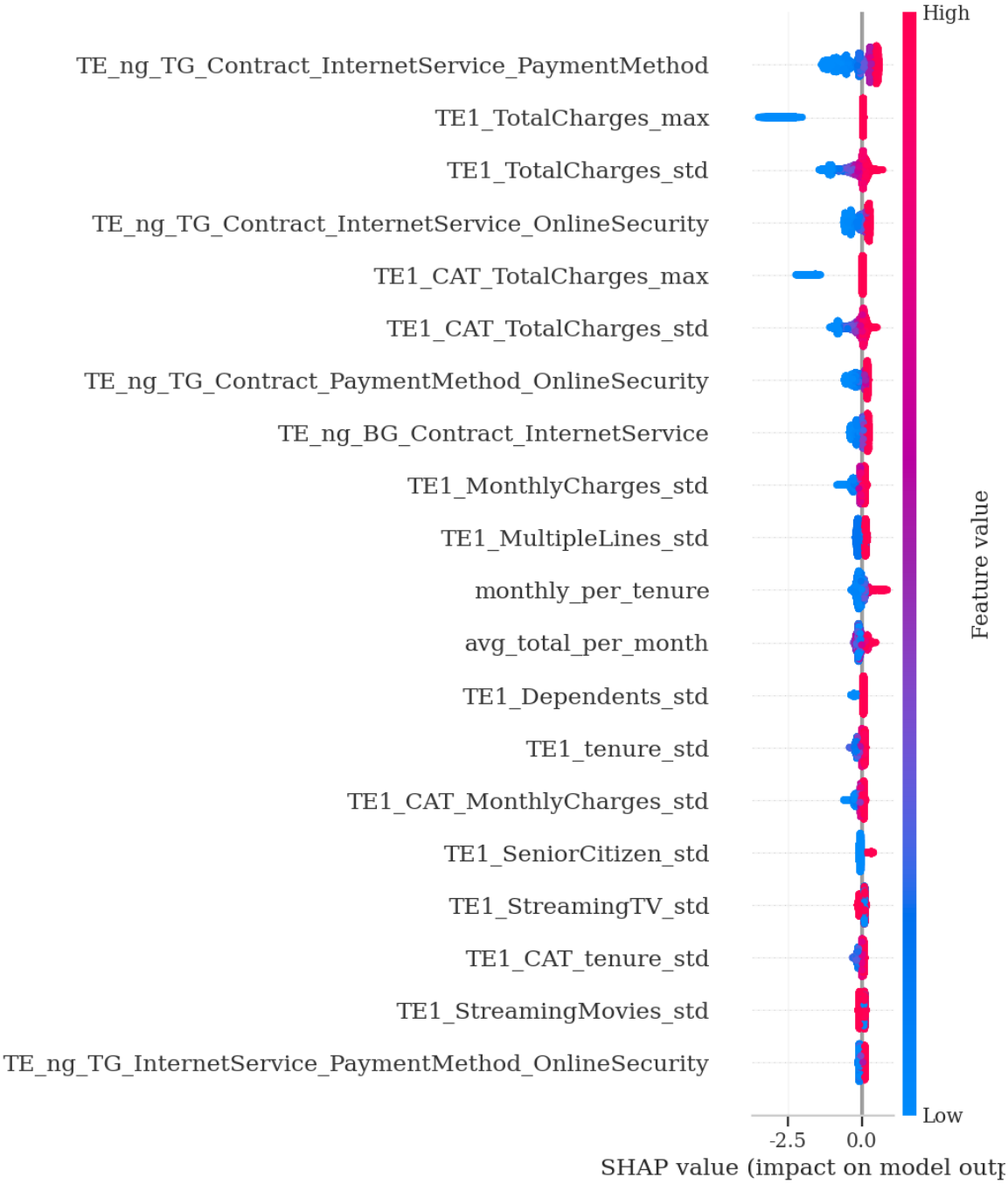
	mean_abs_shap
TE_ng_TG_Contract_InternetService_PaymentMethod	0.530066
TE1_TotalCharges_max	0.369112
TE1_TotalCharges_std	0.321982
TE_ng_TG_Contract_InternetService_OnlineSecurity	0.265316
TE1_CAT_TotalCharges_max	0.247238
TE1_CAT_TotalCharges_std	0.242335
TE_ng_TG_Contract_PaymentMethod_OnlineSecurity	0.233425
TE_ng_BG_Contract_InternetService	0.188321
TE1_MonthlyCharges_std	0.121252
TE1_MultipleLines_std	0.118988
monthly_per_tenure	0.117254
avg_total_per_month	0.115510
TE1_Dependents_std	0.104409
TE1_tenure_std	0.101957
TE1_CAT_MonthlyCharges_std	0.088678
TE1_SeniorCitizen_std	0.088612
TE1_StreamingTV_std	0.087497
TE1_CAT_tenure_std	0.075444
TE1_StreamingMovies_std	0.074691
TE_ng_TG_InternetService_PaymentMethod_OnlineSecurity	0.070517
TE_ng_BG_TechSupport_PaperlessBilling	0.069025
TE_ng_BG_Contract_PaymentMethod	0.065014
TE_ng_BG_Contract_TechSupport	0.061391
TE_ng_BG_Contract_OnlineSecurity	0.060999
ORIG_proba_tenure	0.060460

	mean_signed_shap
TE1_TotalCharges_max	-0.311591
TE1_TotalCharges_std	-0.226661
TE_ng_TG_Contract_InternetService_PaymentMethod	-0.217541
TE1_CAT_TotalCharges_max	-0.209578
TE1_CAT_TotalCharges_std	-0.166938
TE_ng_TG_Contract_PaymentMethod_OnlineSecurity	-0.106774
TE_ng_TG_Contract_InternetService_OnlineSecurity	-0.103124
TE1_MonthlyCharges_std	-0.074320
TE1_tenure_std	-0.056286
TE1_CAT_MonthlyCharges_std	-0.055837
TE_ng_BG_Contract_InternetService	-0.051475
TE1_CAT_tenure_std	-0.041749
TE1_Dependents_std	-0.040867
TE_ng_BG_Contract_PaperlessBilling	-0.037763
TE_ng_BG_TechSupport_PaperlessBilling	-0.034329
TE_ng_BG_Contract_PaymentMethod	-0.031341
monthly_per_tenure	-0.029235
TE_ng_TG_InternetService_PaymentMethod_OnlineSecurity	-0.029133
mc_per_digit	0.027419
avg_total_per_month	-0.025832
ORIG_proba_tenure	-0.025157
TE_ng_BG_PaymentMethod_PaperlessBilling	-0.024866
TE1_SeniorCitizen_std	-0.023626
TE_ng_BG_Contract_TechSupport	-0.023061
TE_ng_BG_Contract_OnlineSecurity	-0.022455

```
saved: analysis_outputs\xgb_02_mean_abs_shap.png
```




```
saved: analysis_outputs\xgb_03_shap_beeswarm.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

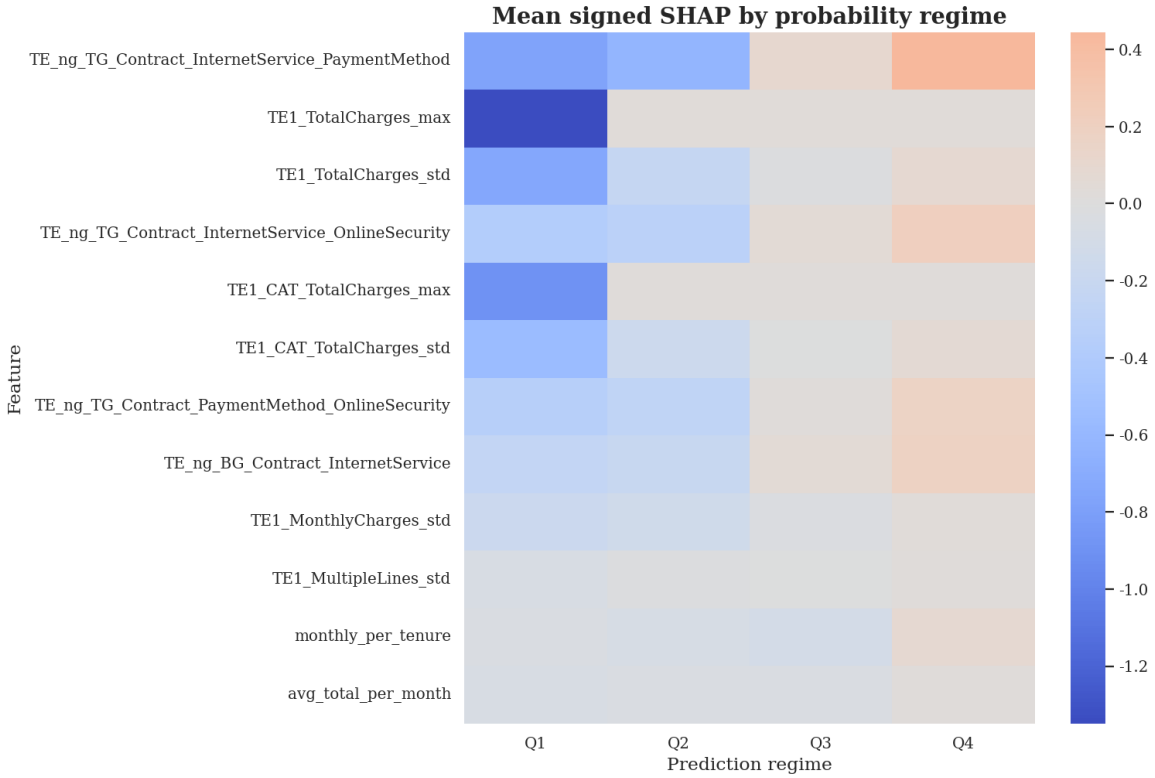
4. Signed SHAP by prediction regime

A feature may be globally important but behave differently in low-risk and high-risk regimes. We therefore compare the mean signed SHAP contributions inside probability bands.

In [7]:
```
shap_df = pd.DataFrame(shap_values, columns=X_shap.columns)
```

```
regime = pd.qcut(xgb_model.predict_proba(X_shap)[:, 1], q=4, labels
shap_df["regime"] = regime.to_numpy()

top20 = mean_abs_shap.head(12).index.tolist()
regime_signed = shap_df.groupby("regime")[top20].mean().T

plt.figure(figsize=(12, 8))
sns.heatmap(regime_signed, cmap="coolwarm", center=0)
plt.title("Mean signed SHAP by probability regime")
plt.xlabel("Prediction regime")
plt.ylabel("Feature")
savefig(cfg, "xgb_04_regime_signed_shap_heatmap.png")
plt.show()

display(regime_signed)
```

```
saved: analysis_outputs\xgb_04_regime_signed_shap_heatmap.png
```



regime	Q1	Q2	
TE_ng_TG_Contract_InternetService_PaymentMethod	-0.780428	-0.629880	0
TE1_TotalCharges_max	-1.348533	0.032095	0
TE1_TotalCharges_std	-0.744716	-0.225837	-(
TE_ng_TG_Contract_InternetService_OnlineSecurity	-0.373592	-0.312692	0
TE1_CAT_TotalCharges_max	-0.904992	0.021995	(
TE1_CAT_TotalCharges_std	-0.559022	-0.163217	-(
TE_ng_TG_Contract_PaymentMethod_OnlineSecurity	-0.344426	-0.272357	0
TE_ng_BG_Contract_InternetService	-0.242659	-0.202735	0
TE1_MonthlyCharges_std	-0.175661	-0.132663	-0
TE1_MultipleLines_std	-0.055214	-0.020275	-(
monthly_per_tenure	-0.042444	-0.067339	-(
avg_total_per_month	-0.058950	-0.039603	-0

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

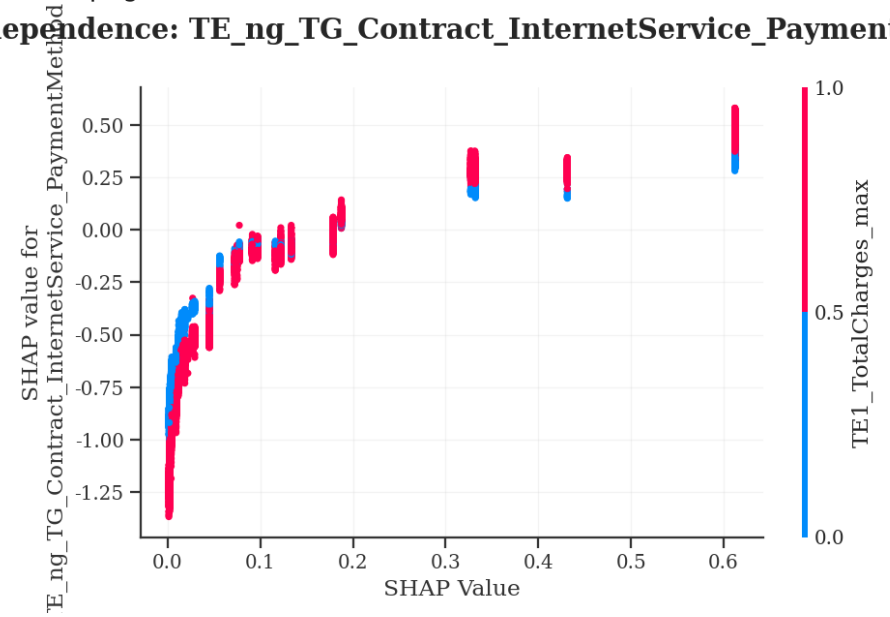
5. Dependence structure and interaction candidates

For top SHAP features, we inspect dependence plots. For trees, visible vertical jumps or terracing are expected because the underlying function is split-based and piecewise constant on the logit scale.

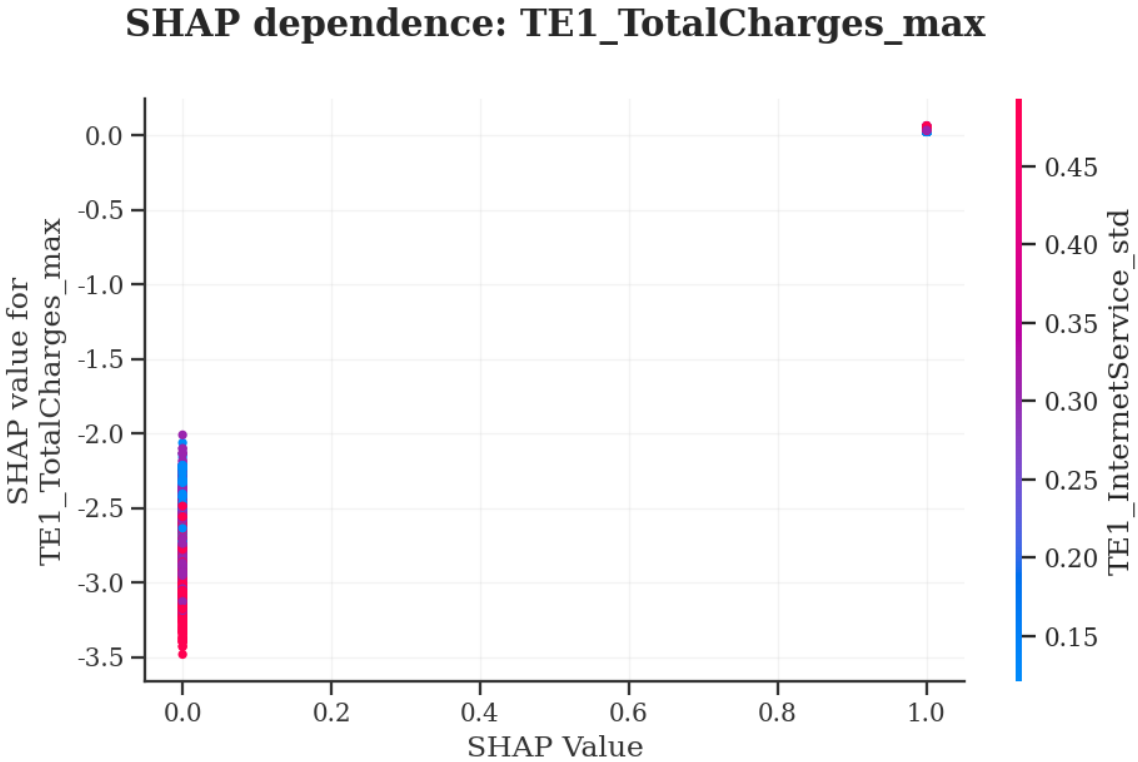
In [9]:
```
top_features = mean_abs_shap.head(8).index.tolist()

for feat in top_features[:6]:
    shap.dependence_plot(
        feat,
        shap_values,
        X_shap,
        interaction_index="auto",
        show=False,
    )
    # Move the title a bit higher up and remove x axis label
    plt.title(f"SHAP dependence: {feat}", pad=28)
    plt.xlabel("SHAP Value")
    fname = f"xgb_dep_{feat.replace('/', '_').replace(' ', '_')}.pn
```

```
    savefig(cfg, fname)
    plt.show()
```

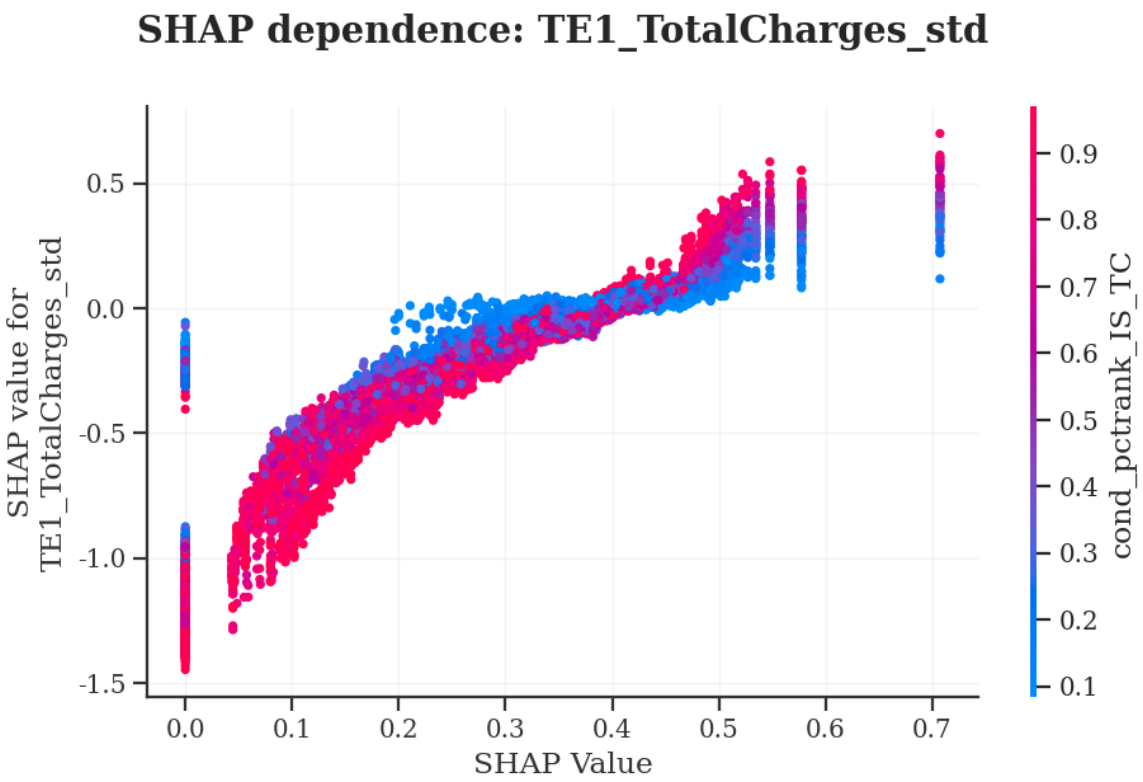
```
saved: analysis_outputs\xgb_dep_TE_ng_TG_Contract_InternetService_Pa
ymentMethod.png
```



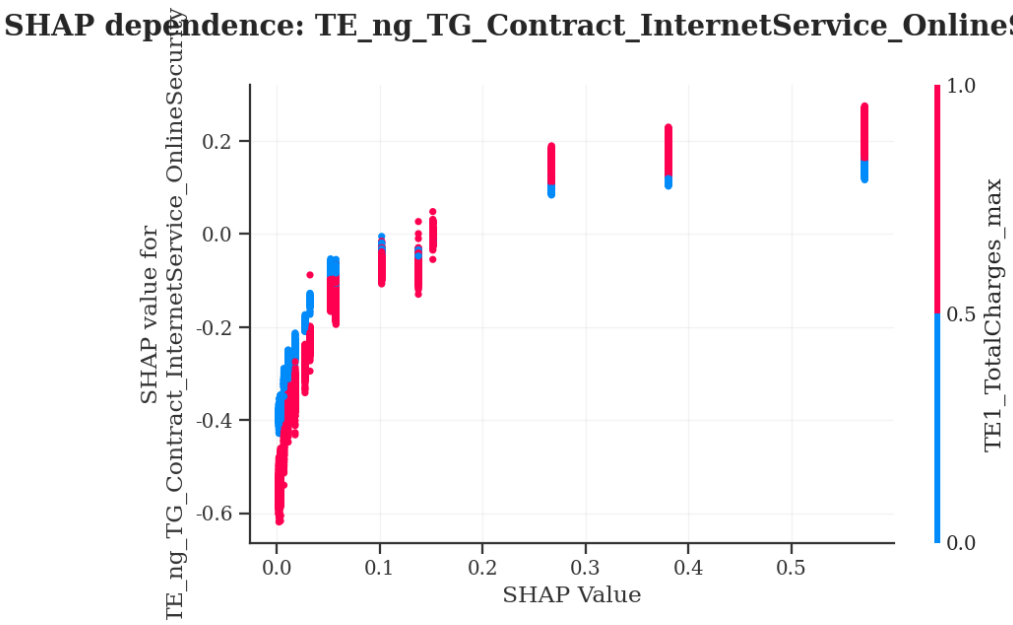

```
saved: analysis_outputs\xgb_dep_TE1_TotalCharges_max.png
```



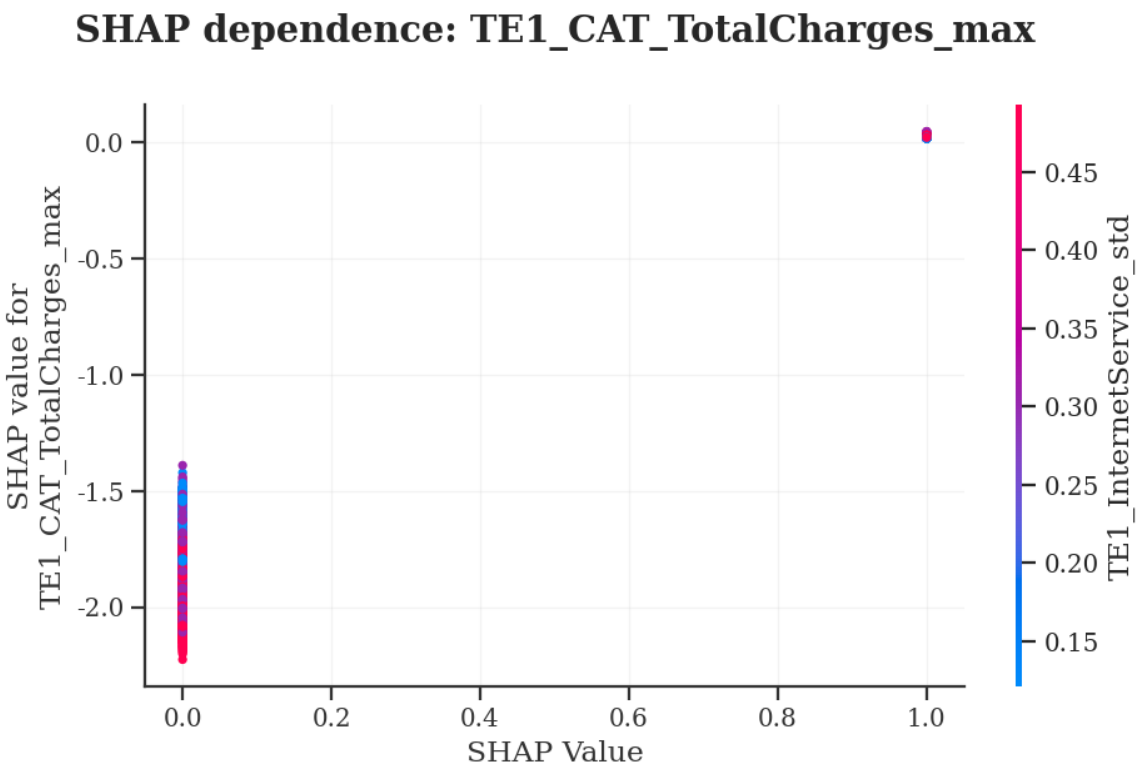

```
saved: analysis_outputs\xgb_dep_TE1_TotalCharges_std.png
```



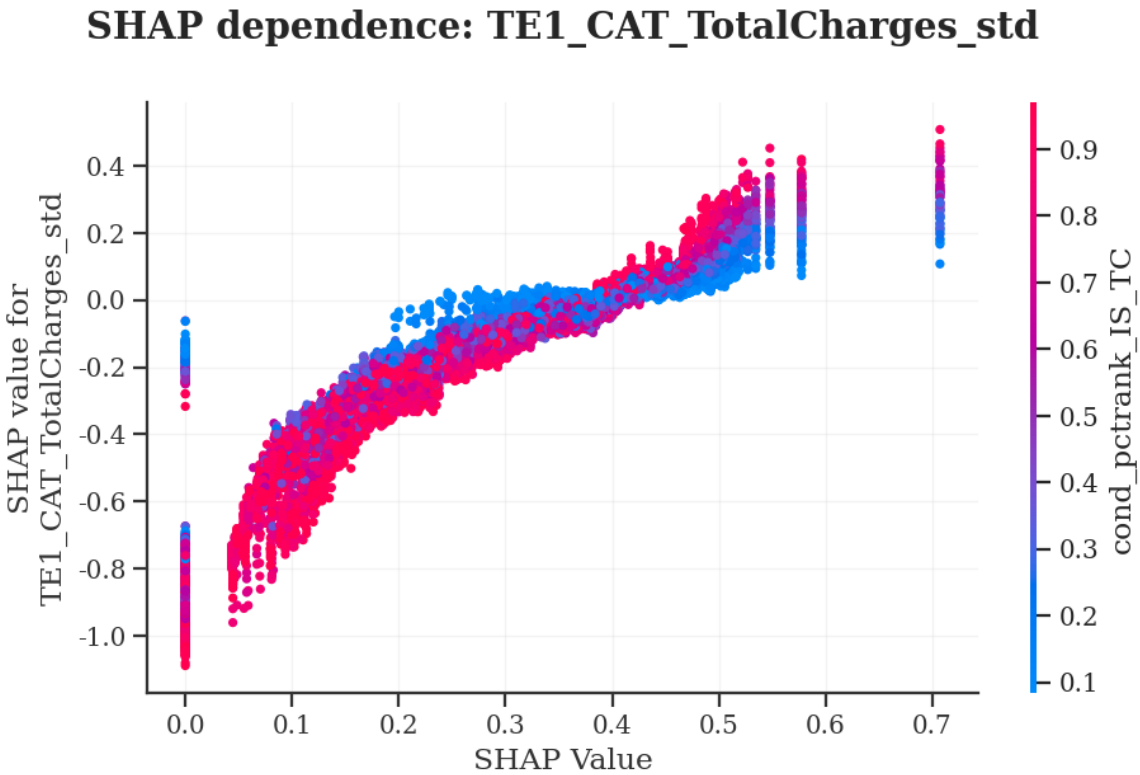

```
saved: analysis_outputs\xgb_dep_TE_ng_TG_Contract_InternetService_On
lineSecurity.png
```




```
saved: analysis_outputs\xgb_dep_TE1_CAT_TotalCharges_max.png
```




```
saved: analysis_outputs\xgb_dep_TE1_CAT_TotalCharges_std.png
```



In [10]:
```
interaction_candidates = []
for feat in top_features[:6]:
    try:
        partners = shap.utils.approximate_interactions(feat, shap_v
        if len(partners) > 1:
            partner = X_shap.columns[partners[1]]
        else:
            partner = X_shap.columns[partners[0]]
        interaction_candidates.append((feat, partner))
    except Exception:
```

```
        pass

interaction_candidates = pd.DataFrame(interaction_candidates, colum
display(interaction_candidates)
```

	feature	approx_partner
0	TE_ng_TG_Contract_InternetService_PaymentMethod	TE1_CAT_TotalCharges_max
1	TE1_TotalCharges_max	ORIG_proba_InternetService
2	TE1_TotalCharges_std	ORIG_proba_tenure
3	TE_ng_TG_Contract_InternetService_OnlineSecurity	TE1_CAT_TotalCharges_max
4	TE1_CAT_TotalCharges_max	ORIG_proba_InternetService
5	TE1_CAT_TotalCharges_std	ORIG_proba_tenure

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

6. Conditional decision surfaces

A tabular decision boundary cannot be visualised directly in full dimension. We therefore inspect two-dimensional conditional slices:

$$(x_i, x_j) \mapsto p_{\text{xgb}}(x_i, x_j, \tilde{x}_{-ij}),$$

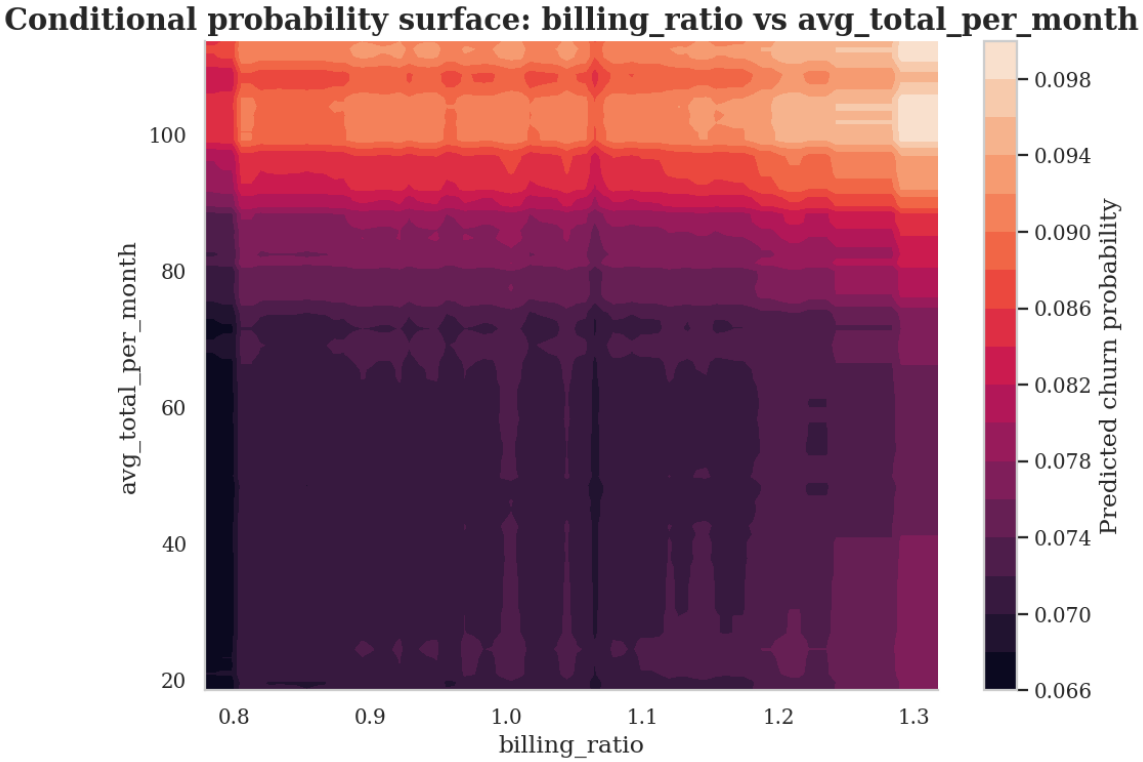
where all non-focal coordinates are fixed at a reference profile.

The resulting surfaces should reveal staircase-like geometry if the tree ensemble is using threshold partitions sharply.

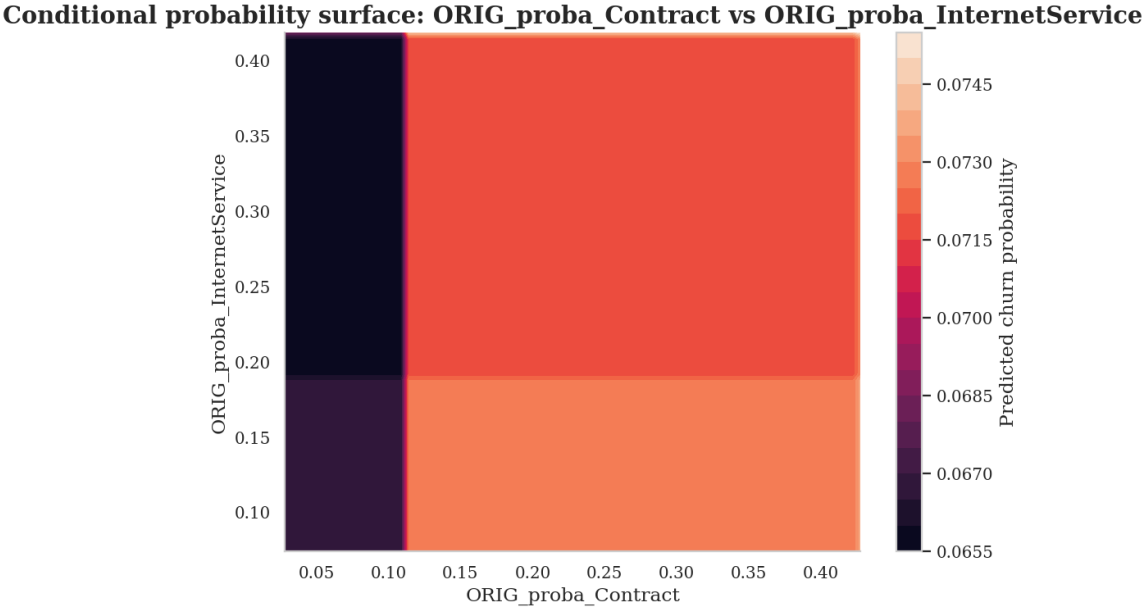
In [11]:
```
def make_reference_row(X):
    ref = {}
    for c in X.columns:
        s = X[c]
        if pd.api.types.is_numeric_dtype(s):
            ref[c] = float(s.median())
        else:
            mode = s.mode(dropna=True)
            ref[c] = mode.iloc[0] if len(mode) else s.iloc[0]
    return pd.Series(ref)

def plot_xgb_surface(model, X_ref, feature_x, feature_y, q_low=0.02
    row = make_reference_row(X_ref)
    xs = np.linspace(X_ref[feature_x].quantile(q_low), X_ref[featur
    ys = np.linspace(X_ref[feature_y].quantile(q_low), X_ref[featur
```

```
    mesh = []
    for yv in ys:
        tmp = pd.DataFrame([row] * len(xs))
        tmp[feature_x] = xs
        tmp[feature_y] = yv
        mesh.append(tmp)
    mesh = pd.concat(mesh, axis=0, ignore_index=True)
    z = model.predict_proba(mesh[X_ref.columns])[:, 1].reshape(len(

    fig, ax = plt.subplots(figsize=(8, 6))
    cf = ax.contourf(xs, ys, z, levels=20)
    ax.contour(xs, ys, z, levels=[0.5], linewidths=2)
    ax.set_xlabel(feature_x)
    ax.set_ylabel(feature_y)
    ax.set_title(f"Conditional probability surface: {feature_x} vs
    plt.colorbar(cf, ax=ax, label="Predicted churn probability")
    return fig, ax

surface_pairs = []
for a, b in [
    ("tenure", "MonthlyCharges"),
    ("billing_ratio", "avg_total_per_month"),
    ("ORIG_proba_Contract", "ORIG_proba_InternetService"),
    ("tenure_mod12", "MonthlyCharges"),
]:
    if a in X_train_aligned.columns and b in X_train_aligned.column
        surface_pairs.append((a, b))

for a, b in surface_pairs:
    fig, ax = plot_xgb_surface(xgb_model, X_train_aligned.iloc[:n_c
    savefig(cfg, f"xgb_surface_{a}_{b}.png".replace("/", "_").repla
    plt.show()
```

```
saved: analysis_outputs\xgb_surface_billing_ratio_avg_total_per_mont
h.png
```




```
saved: analysis_outputs\xgb_surface_ORIG_proba_Contract_ORIG_proba_I
nternetService.png
```



Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

8. Error-specific explanation

We now compare SHAP structures for:

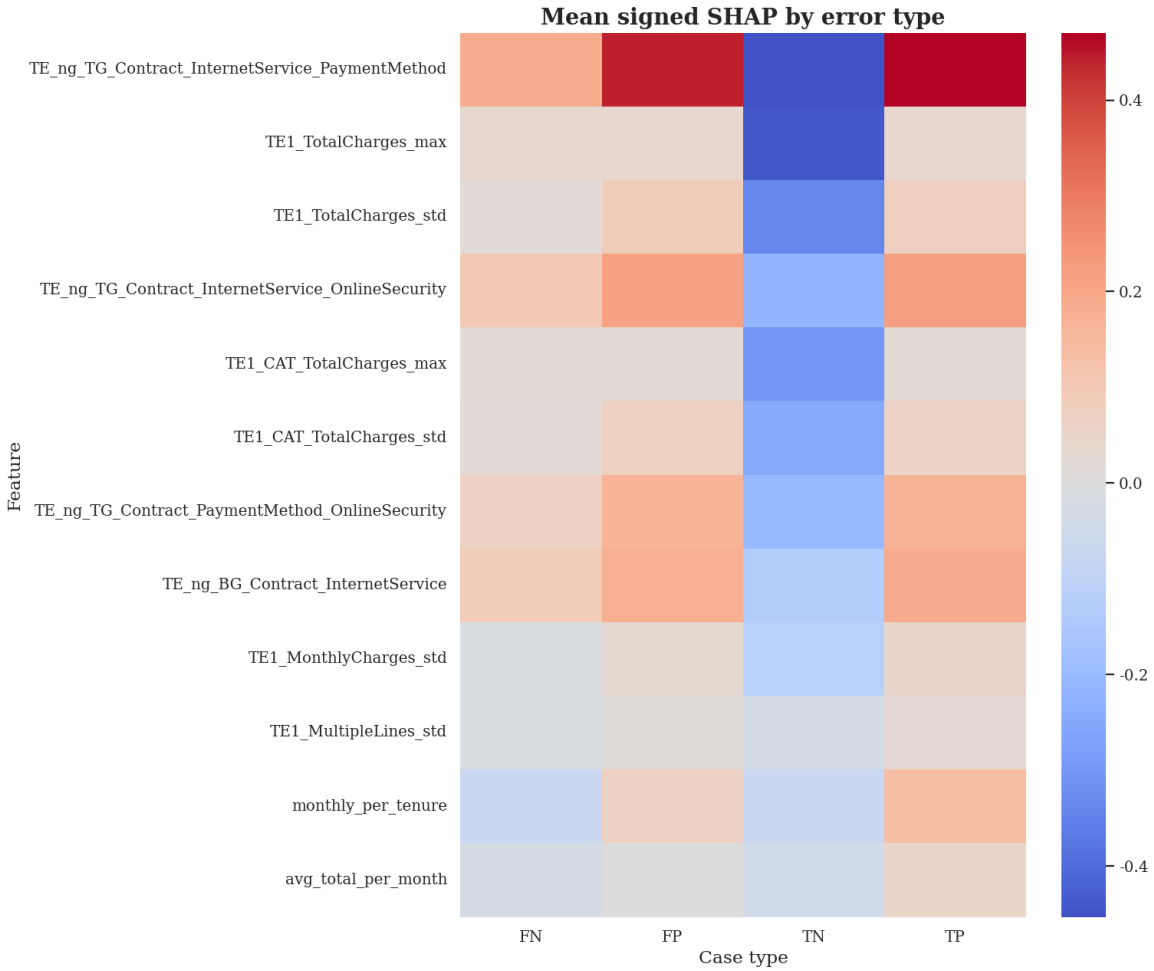
- false positives,

- false negatives,
- true positives,
- true negatives.

This isolates where the model's inductive bias is helping and where it is misallocating risk.

In [13]:
```
shap_analysis = analysis_train.loc[shap_idx, [cfg.id_col, "y_true",
group_means = shap_analysis.groupby("case_type")[top20].mean().T

plt.figure(figsize=(12, 10))
sns.heatmap(group_means, cmap="coolwarm", center=0)
plt.title("Mean signed SHAP by error type")
plt.xlabel("Case type")
plt.ylabel("Feature")
savefig(cfg, "xgb_05_signed_shap_by_case_type.png")
plt.show()

display(group_means)
```

```
saved: analysis_outputs\xgb_05_signed_shap_by_case_type.png
```



case_type	FN	FP	
TE_ng_TG_Contract_InternetService_PaymentMethod	0.187166	0.441763	-0.
TE1_TotalCharges_max	0.034215	0.035932	-0.
TE1_TotalCharges_std	0.016664	0.090646	-0.
TE_ng_TG_Contract_InternetService_OnlineSecurity	0.102045	0.212919	-0.
TE1_CAT_TotalCharges_max	0.021921	0.022577	-0.
TE1_CAT_TotalCharges_std	0.016465	0.071195	-0.
TE_ng_TG_Contract_PaymentMethod_OnlineSecurity	0.065052	0.165380	-0.
TE_ng_BG_Contract_InternetService	0.088950	0.180518	-0.
TE1_MonthlyCharges_std	-0.011059	0.032678	-0
TE1_MultipleLines_std	-0.009929	0.011905	-0.
monthly_per_tenure	-0.069810	0.067704	-0.
avg_total_per_month	-0.032325	0.003988	-0.

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

9. Local explanations

For representative cases, we inspect full SHAP decompositions.

The local decomposition is

$$f_{\text{xgb}}(x^{(i)}) = \phi_0 + \sum_{j=1}^{p} \phi_j(x^{(i)}).$$

We deliberately choose extreme true positives, extreme true negatives, a false positive, and a false negative where available.

In [14]:
```
candidate_indices = []

tp_idx = analysis_train.query("case_type == 'TP'").sort_values("xgb
tn_idx = analysis_train.query("case_type == 'TN'").sort_values("xgb
fp_idx = analysis_train.query("case_type == 'FP'").sort_values("xgb
fn_idx = analysis_train.query("case_type == 'FN'").sort_values("xgb

for arr in [tp_idx, tn_idx, fp_idx, fn_idx]:
```

```
    if len(arr):
        candidate_indices.append(int(arr[0]))

candidate_indices = [i for i in candidate_indices if i in set(shap_
candidate_positions = [int(np.where(shap_idx == i)[0][0]) for i in

for pos in candidate_positions:
    row_meta = analysis_train.loc[shap_idx[pos], ["y_true", "xgb_pr
    expl = shap.Explanation(
        values=shap_values[pos],
        base_values=expected_value_scalar,
        data=X_shap.iloc[pos].values,
        feature_names=X_shap.columns.tolist(),
    )
    print(row_meta.to_dict())
    shap.plots.waterfall(expl, max_display=20, show=False)
    savefig(cfg, f"xgb_waterfall_row_{int(shap_idx[pos])}.png")
    plt.show()
```

Mathematical framing

In this section, I study the model as a function $\hat{p}(x)$ mapping inputs to probabilities. I interpret each computation as evaluating properties of this function such as gradients, perturbations, or expectations.

10. Export tables

The notebook writes the central outputs needed for a report:

- fitted train/test predictions,
- decile table,
- SHAP rankings,
- regime-specific SHAP summaries,
- case-type SHAP summaries.

In [15]:
```
outdir = Path(cfg.output_dir)

analysis_train.to_csv(outdir / "xgb_analysis_train.csv", index=Fals
analysis_test.to_csv(outdir / "xgb_analysis_test.csv", index=False)
dec.to_csv(outdir / "xgb_deciles.csv", index=False)
mean_abs_shap.rename("mean_abs_shap").to_csv(outdir / "xgb_mean_abs
mean_signed_shap.rename("mean_signed_shap").to_csv(outdir / "xgb_me
regime_signed.to_csv(outdir / "xgb_regime_signed_shap.csv")
group_means.to_csv(outdir / "xgb_case_type_signed_shap.csv")
interaction_candidates.to_csv(outdir / "xgb_interaction_candidates.

summary = {
    "train_auc_competition_only": float(roc_auc_score(y, xgb_comp_t
    "train_logloss_competition_only": float(log_loss(y, xgb_comp_tr
    "train_brier_competition_only": float(brier_score_loss(y, xgb_c
    "n_saved_features": int(len(feature_cols_saved)),
```

```
    "n_runtime_rows_train": int(X_train_aligned.shape[0]),
    "n_runtime_rows_test": int(X_test_aligned.shape[0]),
    "n_runtime_cols": int(X_train_aligned.shape[1]),
    "top10_features_mean_abs_shap": mean_abs_shap.head(10).to_dict(
}

(outdir / "xgb_xai_summary.json").write_text(json.dumps(summary, in
summary
```

Out[15]:

```
{'train_auc_competition_only': 0.9334367672728091,
 'train_logloss_competition_only': 0.26805529743260703,
 'train_brier_competition_only': 0.0859934464097023,
 'n_saved_features': 161,
 'n_runtime_rows_train': 601237,
 'n_runtime_rows_test': 254655,
 'n_runtime_cols': 161,
 'top10_features_mean_abs_shap': {'TE_ng_TG_Contract_InternetServi
ce_PaymentMethod': 0.5300655364990234,
  'TE1_TotalCharges_max': 0.3691123425960541,
  'TE1_TotalCharges_std': 0.3219815790653229,
  'TE_ng_TG_Contract_InternetService_OnlineSecurity': 0.2653157413
005829,
  'TE1_CAT_TotalCharges_max': 0.2472381889820099,
  'TE1_CAT_TotalCharges_std': 0.24233508110046387,
  'TE_ng_TG_Contract_PaymentMethod_OnlineSecurity': 0.233424514532
08923,
  'TE_ng_BG_Contract_InternetService': 0.18832050263881683,
  'TE1_MonthlyCharges_std': 0.12125235795974731,
  'TE1_MultipleLines_std': 0.11898791790008545}}
```

In [ ]: